

Yael Eisenberg

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EDUCATION

Cornell University

2019 – 2025

Ph.D. in Mathematics GPA: 3.97/4.0

Ithaca, NY

- **Dissertation:** [Counting Short Vectors, Combining Easy Problems, and Observing Numbers](#) Advisor: [Noah Stephens-Davidowitz](#)
- Researched lattice geometry, cryptography, and numerical cognition with collaborators at Cornell, Tel Aviv University, UCLA, and the Washington Corrections Center; proved a [tight reverse Minkowski inequality for the Epstein zeta function](#).
- **Four Research Results:** [Proc. AMS](#), [Serdica Math. J.](#), [arXiv](#), [IACR ePrint](#)
- **Awards:** [NSF Graduate Research Fellowship](#); [Cornell University Fellowship](#); [AWM Emerging Mathematicians Travel Grant](#)

Yeshiva University

2015 – 2020

B.A. (2019) / M.A. (2020) in Mathematics GPA: 3.98/4.0

New York, NY

- **Master's Thesis:** [Restricted Planar Three-Body Problem, Convexity, and Global Surface of Section](#) Advisor: [Marian Gidea](#)
- **Awards:** S. Daniel Abraham Honors Program; Gilda R. S. Silverman Memorial Award for Excellence in Mathematics; Kressel Research Scholarship

EXPERIENCE

Mercor Intelligence

2025 – Present

Math Subject Matter Expert & Reviewer

Remote

- Improve model performance in advanced mathematics for a leading AI lab; review and quality-check other mathematicians' submissions, enforcing rigorous standards before delivery.

Two Six Technologies

Summers 2023, 2024

Research Intern (2×), Formal Verification

Arlington, VA / Remote

- Verified correctness and security properties of cryptographic algorithms with Cryptol and SMT solvers (Z3, CVC5), including the NTT in the post-quantum Dilithium scheme and Montgomery multiplication.
- Presented SMT-based proofs to the research group, validating proposed cryptographic architectures.

NASA Kennedy Space Center

Spring 2019

Launch Vehicle Control Intern

Cape Canaveral, FL

- Modeled launch vehicle control systems in Python, MATLAB, and Simulink; analyzed Monte Carlo simulation data to refine controller designs and presented results to NASA aerospace engineers.

TEACHING, LEADERSHIP & OUTREACH

National Museum of Mathematics (MoMath)

2025 – Present

MOST Fellow (Mathematics Outreach Seminar & Training)

New York, NY

- Selected nationally for intensive communication training, including workshops at the Peoples Improv Theater.
- Design and deliver three interactive talks for middle-school girls, including one on tessellation and symmetry at MoMath; panelist for MoMath Women's "Day in the Life of a Mathematician."

Cornell University

2020 – 2025

Instructor of Record (2025); Teaching Assistant (2020 – 2023)

Ithaca, NY

- Taught 175+ undergraduate students in Calculus, Linear Algebra, Probability, and Matrix Groups.

Association for Women in Mathematics (AWM), Cornell Chapter

2019 – 2022

Mentoring Program Officer

Ithaca, NY

- Matched undergraduate women with graduate mentors based on mathematical and personal interests, and ran 3–5 professional development events per semester, including career panels and imposter-syndrome workshops.

SELECTED PROJECTS

Forecasting NYC 311 Rat Sightings [\[Link\]](#)

Spring 2026

Erdős Institute Data Science Boot Camp

- Forecast daily 311 rat sightings 14 days ahead (citywide and by borough) using weather and calendar features with 26-fold cross-validation; a hybrid Prophet + XGBoost model cut citywide RMSE 37% on the holdout set (17.57 → 10.98) versus a seasonal baseline.

SKILLS

- **Technical:** Python (pandas, NumPy, scikit-learn, Prophet, XGBoost), SageMath, Mathematica, MATLAB, Cryptol, Z3, CVC5, $\text{\LaTeX}$
- **Languages:** English (native), Hebrew (fluent)
- **Interests:** Unicycling, violin, swimming, martial arts, hiking, 3D printing, laser cutting, volunteer EMT, sewing, crochet